

Digital Sound

I2S carries audio as a clocked bit stream — ESP32 to DAC to amp to speaker. Clean sound is a grounding lesson you can hear.

ESP32-009 · ~50 MIN · ADVANCED

1. The one idea

I2S is a dedicated digital-audio bus: three clocked lines carry sound as a precise bit stream. Send audio out of the ESP32 to a DAC, through a small amp, to a speaker — and hear, directly, what clean grounding and decoupling do.

2. Why it matters

Every quality audio part — DACs, ADCs, MEMS mics — speaks I2S. And audio is a merciless teacher of analog discipline: a ground loop or a missing decoupling cap doesn't crash the program, it **hisses and whines**, audibly. You hear your layout mistakes, which makes this the best lesson on the bench for grounding done right.

3. Parts & wiring

- ESP32 dev board
- PCM5102 I2S DAC module
- PAM8403 class-D amplifier
- 4Ω 3W speaker

I2S line	ESP32	PCM5102	Carries
BCLK	GPIO26	BCK	the bit clock
LRCK	GPIO25	LRCK	left/right word select
DOUT	GPIO22	DIN	the audio data

*Module setup and the grounding that matters. On the PCM5102, tie **SCK to GND** (it uses its internal PLL) and **XSMT to 3V3** (un-mute), or it stays silent. Then the analog rules: a single **star ground** — digital and analog returns meet at one point, never in a loop. Give the DAC's analog supply its own **0.1μF + 10μF**. **AC-couple** the amp input (series cap) so no DC offset thumps the speaker, and never tie the speaker return into the digital ground plane.*

4. Predict (do this before uploading)

- Three I2S lines: which is the bit clock, which the left/right select, which the data?
- A ground loop between the DAC and the amp — will it sound like silence, or a tone?
- What do you expect a *missing* supply decoupling cap to sound like?

5. Build it — a clean tone over I2S

```
#include <driver/i2s.h>
#include <math.h>

const int SR = 44100;
const i2s_port_t PORT = I2S_NUM_0;

void setup() {
    i2s_config_t cfg = {
        .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_TX),
        .sample_rate = SR,
        .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
        .channel_format = I2S_CHANNEL_FMT_RIGHT_LEFT,
        .communication_format = I2S_COMM_FORMAT_STAND_I2S,
        .intr_alloc_flags = 0,
        .dma_buf_count = 8,
        .dma_buf_len = 64
    };
    i2s_pin_config_t pins = {
        .bck_io_num = 26, .ws_io_num = 25,
        .data_out_num = 22, .data_in_num = I2S_PIN_NO_CHANGE
    };
    i2s_driver_install(PORT, &cfg, 0, NULL);
    i2s_set_pin(PORT, &pins);
}

void loop() {
    static float phase = 0;
    int16_t buf[256];
    for (int n = 0; n < 256; n += 2) {
        int16_t s = (int16_t)(sin(phase) * 8000);    // 440 Hz, modest amplitude
        buf[n] = s; buf[n + 1] = s;                // left + right
        phase += 2 * PI * 440 / SR;
        if (phase > 2 * PI) phase -= 2 * PI;
    }
    size_t written;
```

```
i2s_write(PORT, buf, sizeof(buf), &written, portMAX_DELAY);  
}
```

The new idea: I2S streams samples on a hardware-clocked bus via DMA, so the timing is exact and the CPU just fills buffers. You generate a 440 Hz sine, the DAC makes it analog, the amp drives the speaker.

6. Test against your prediction

You should hear a clean, steady A-440 tone. Now go looking for trouble: run the speaker return into the digital ground and listen for hum; pull the analog decoupling and listen for hiss/whine. Fix the star ground and the bypass caps and the background goes silent. Did the ground loop sound the way you predicted?

7. What goes wrong (pre-loaded debugging)

- **No sound at all** → PCM5102 SCK not grounded, XSMT left low (muted), or an I2S pin miswired.
- **Hum or whine under the tone** → a ground loop. Bring digital and analog grounds to one star point.
- **Hiss / digital "mosquito" noise** → missing supply decoupling, or I2S traces run long next to noisy lines. Bypass and shorten.
- **Distortion / thump** → no AC coupling on the amp input (DC offset), or the amplitude/gain is clipping. Add the series cap; back off level.
- **Channels swapped or warbling** → LRCK polarity / word-select timing wrong; check the I2S format and pin mapping.

8. Checkpoint

You moved digital audio over I2S and heard, in real time, exactly what grounding and decoupling do to a signal. Sound is analog truth: it rewards a clean star ground and honest bypassing and punishes a sloppy layout out loud. That ear is worth as much as any meter.

9. Push further

- Flip it around: read the **INMP441** I2S mic *in*, and measure live sound level.
- Pipe that level to the WS2812B strip (Course 6) for a sound-reactive light.
- Run an FFT on the mic stream and show a spectrum on the OLED.

